THE UNIVERSITY OF WESTERN AUSTRALIA

EXAMINATION PAPER CHECKLIST
CITS2200 Data Structures and Algorithms

This page is to remain part of your examination file and is to be submitted with your Examination Cover Sheet and content. This page is for information only and will not be printed.

☐ Questions for the examination commence on the page following the Examination Cover Sheet. The examination paper must have page numbers and should state e.g. Page 1 of x.

☐ Ensure type of examination is correct 'permitted materials or no permitted materials'

- **Permitted materials**–Specific materials are permitted. Permitted material selected from the 'Materials Permitted' and 'Other Materials Permitted' area of the online examination request portal will be printed on the Examination Cover Sheet under the headings 'Supplied by the Student'.
- **No permitted materials**-no material is permitted.

☐ All pages, sections and questions are numbered sequentially i.e., pages, Part A, B, C, … etc. Questions 1, 2, 3, … Subsections to question numbering is to be consistent throughout the examination paper i.e. (a), (b), (c), …

☐ General instructions have been entered in the 'Instructions to Students' area of the online exam request portal and are reflected on the Examination Cover Sheet. Instructions have also been clearly communicated to students e.g., Answer Part A in the answer book provided and Part B on the examination paper.

☐ 'END OF EXAMINATION PAPER' is to be stated on the last page of the examination paper. END OF ATTACHMENT' is to be stated on the last page of the attachment where applicable.

☐ The examination paper is of a high-quality readable format, e.g., consistent formatting no blurred text, images are clear and printable.

☐ The document has been saved as a PDF and retains the downloaded unit code. I have also made no adjustment to the template which is A4.

☐ Where relevant the coversheet has been duplicated and an attachment has been included with a separate cover sheet at the end of the examination paper. All attachments will be printed as separate documents. If adding an attachment use a section break on the attachment cover page to restart page numbering.

☐ The level of distinctiveness for main round exams is at least 50% from the exam in the previous teaching period. The level of distinctiveness for Supplementary and Deferred exams is at least 25% from the main round exams.

☐ If this is an anonymous paper, Family Name, Given Name and Signature has been deleted from the cover sheet(s).

The examination paper has been proofread, the above checks completed, and approved for submission.

	NAME OR ELECTRONIC SIGNATURE	DATE
UNIT COORDINATOR (EXAMINER)	Amitava Datta	29/4/2025
CO-EXAMINER (OPTIONAL)		

DESK No.

FAMILY NAME: ______________________

GIVEN NAMES: ______________________

SIGNATURE: ______________________

STUDENT NUMBER:

Semester 1, 2025 EXAMINATIONS

School of Physics, Mathematics and Computing

CITS2200

Data Structures and Algorithms

Examination Duration: 2 hours

This is an examination WITH permitted materials

Provided by the University
1 x 18 Page Answer Booklet

Supplied by the Student

Calculator
UWA approved calculators with stickers are permitted

Instructions to Students

1. This paper has four pages including this cover page.
2. There are five questions, each with two parts.
3. Total marks for this paper is 50.
4. Students should answer all questions

Examination Cover Sheet

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Q1.

(a) Explain the `Merge Sort` algorithm in your own words. Explain its complexity when the size of the input is n.

(5 marks)

(b) Explain why the `Merge Sort` algorithm may not be efficient practically for large inputs. Explain a strategy how `Merge Sort` can be combined with `Insertion Sort` to design a more efficient sorting algorithm for large inputs. Explain why your strategy is more efficient in practice.

(5 marks)

Q2.

(a) You are given an array A of n numbers sorted in ascending order. You are also given an array B of m numbers that is unsorted. Assume all the numbers in the two arrays are distinct (no number appears more than once in any of the two arrays). Design an algorithm that finds the k-th smallest number in $A \cup B$ (among all the numbers in the arrays A and B). Explain your algorithm in English and explain its complexity in terms of n, m and k.

(5 marks)

(b) You are given a sorted array A of n integers in descending order, and another integer q. Explain two algorithms that will either find two integers r and s in A such that $r + s = q$, or report that no such pair r, s exists. Your first algorithm should run in $O(n \log n)$ time, and your second algorithm should run in $O(n)$ time.

(5 marks)

Q3.

(a) Arrange the following asymptotic complexities in ascending order:
$O(2^n), O(n \log \log n), O(n \log^2 n), O(n^2 \log n), O(2^{\log n})$. Explain the reason behind your ordering.

(5 marks)

(b) We add a new operation `multiPop(i)` to the stack ADT that pops i items from a stack. Explain clearly why the amortized complexity of the `multiPop(i)` operation is $O(1)$.

(5 marks)

Q4.

(a) Explain clearly the `inorder`, `preorder` and `postorder` traversals of a tree. What are the complexities of these three traversal algorithms, if there are n internal nodes in the tree?

(5 marks)

(b) Explain the `Heap Sort` algorithm clearly. What is the complexity of this algorithm? Explain your answer clearly.

(5 marks)

Q5.

(a) Explain clearly Prim's `minimum spanning tree` algorithm. Explain briefly how it is implemented. What is the complexity of this algorithm?

(5 marks)

(b) Explain clearly Dijkstra's `single-source shortest path` algorithm for a graph with non-negative edge weights. Explain how the `priority queue` data structure is used for implementing this algorithm, and how the priority queue is updated in each iteration of the algorithm. Explain the complexity of this algorithm.

(5 marks)

END OF EXAMINATION PAPER